Question One

We are given two planes in parametric form, Π_1 and Π_2

$$\Pi_1 : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

where $\lambda_1,\ \lambda_2 \in \mathbb{R}$, and

$$\Pi_2 : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \mu_1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \mu_2 \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix}$$

where $\mu_1,\ \mu_2 \in \mathbb{R}$.

a) We are trying to find two vectors $\overrightarrow{n_1}$ and $\overrightarrow{n_2}$ that are normal to Π_1 and Π_2. To do this we take the cross product of the vector within each plane, as it gives the normal vector.

For Π_1

$$\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix},$$

And for Π_2

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix}.$$

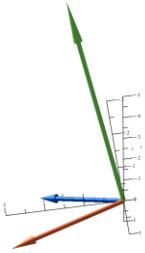
Performing the calculations in Maple also confirm the result, as expected. Furthermore, plotting the planes helps visualise the result. The graphs of Π_1, and Π_2 are shown below respectively.

```
u := ⟨2, 1, −1⟩ :
v := ⟨3, 0, 1⟩ :
cross_product := u&xv;
```

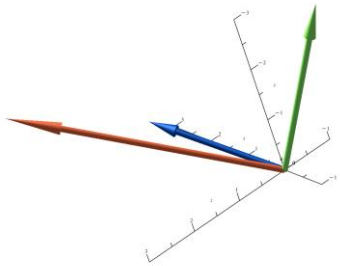
$$cross_product := \begin{bmatrix} 1 \\ -5 \\ -3 \end{bmatrix} \quad (1)$$

```
u_2 := ⟨2, −1, 1⟩ :
v_2 := ⟨3, −2, 3⟩ :
cross_product := u_2&xv_2;
```

$$cross_product := \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix} \quad (2)$$



In green, the cross product of the blue vector crossed into the red vector.



In green, the cross product of the blue vector crossed into the red vector.

The second part of the question ask how we can tell, without performing any calculations that Π_1 and Π_2 must intersect in a line. Two planes in $\mathbb{R}^3$ must intersect in a line if their normal vectors are not parallel. In other words, if the normals of two planes are not scalar multiples of each other, the planes are not parallel and must intersect. This is analogous to how two non-parallel line in $\mathbb{R}^2$ must intersect at a point.

b) To find the Cartesian equation from parametric form, we need:

- a vector normal to the plane, and
- a point on the plane and the normal.

This is because the Cartesian equation for a plane is simply an expansion of the point normal form of a plane. Below shows the working just for Π_1, but the same steps are done for Π_2.

Calculated previously, the normal of the plane is

$$\overrightarrow{n_1} = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix}.$$

By setting λ_1 and λ_2 to zero in the parametric equation a point on the plane is

$$\overrightarrow{r_0} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}.$$

Using the point normal form

$$\overrightarrow{n_1} \cdot (\vec{x} - \overrightarrow{r_0}) = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix} \cdot \left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} x_1 - 3 \\ x_2 - 2 \\ x_3 - 0 \end{pmatrix} = 0.$$

Turning this into Cartesian equation is just expanding the brackets and simplifying

$$x_1 - 5x_2 - 3x_3 = -7.$$

Turning x_1, x_2 and x_3 into x, y and z respectively give

$$\Pi_1 : x - 5y - 3z = -7,$$

$$\Pi_2 : -x - 3y - z = -5.$$

Verifying this equation in Maple give the same result.

```
p := ⟨3, 2, 0⟩ :   # A point on the plane
n := ⟨1, −5, −3⟩ : # This is the normal to the plane calculated before
x := x : y := y : z := z :
DotProduct(n, ⟨x, y, z⟩ − p);
```

$$x + 7 - 5y - 3z \qquad (1)$$

```
p := ⟨3, 0, 2⟩ :   # A point on the plane
n := ⟨−1, −3, −1⟩ : # This is the normal to the plane calculated before
x := x : y := y : z := z :
DotProduct(n, ⟨x, y, z⟩ − p);
```

$$-x + 5 - 3y - z \qquad (2)$$

c) We are asked to find the parametric vector form L for the line of intersection between the two planes Π_1 and Π_2, given in Cartesian form

$$\Pi_1: x - 5y - 3\omega = -7,$$

$$\Pi_2: -x - 3y - \omega = -5.$$

It explicitly states that we must assign one variable (i.e. x_1) to a parameter ω, and then expressing the other two variable in terms of ω. Below shows the working when turning z into ω

$$5y + 3\omega - 7 = x,$$

$$y = \frac{3 - \omega}{2},$$

$$x = \frac{1 + \omega}{2}.$$

Since we originally set $z = \omega$, the final parametric form is

$$L: x = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \\ 0 \end{pmatrix} + \omega \begin{pmatrix} \frac{1}{2} \\ \frac{-1}{2} \\ 1 \end{pmatrix},$$

where $\omega \in \mathbb{R}$.

d) For the second method we are asked to substitute the parametric form of Π_2 into the Cartesian form of Π_1, and once again find the parametric vector form of the line of intersection L.

Cartesian equation for Π_1 found in part b

$$x - 5y - 3z = -7.$$

Parametric equation for Π_2

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \mu_1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \mu_2 \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix}.$$

Let's make x_1, $x_2$and x_3 into x, y and z respectively and find the expression for each

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 + 2\mu_1 + 3\mu_2 \\ 0 - \mu_1 - 2\mu_2 \\ 2 + \mu_1 + 3\mu_2 \end{pmatrix}.$$

Subbing this expression into the Cartesian equation for Π_1

$$\mu_1 + \mu_2 = -1.$$

Let $\mu_1 = \omega$ so $\mu_2 = -1 - \omega$ and subbing this back into Π_2

$$3 + 2\omega + 3(-1 - \omega) = -\omega,$$
$$0 - \omega - 2(-1 - \omega) = 2 + \omega,$$
$$2 + \omega + 3(-1 - \omega) = -1 - 2\omega.$$

Hence the parametric form is

$$L: x = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} + \omega \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}.$$

e) The parametric form in part (c) and part (d) appear different; however, they do represent the same line L because both their vectors are scalar multiple of each other. There can be infinitely many vectors and point to represent the same line. In this example (d) is a scalar multiple of negative two in reference to (c) as

$$-2 \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{-1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix},$$

where $\omega \in \mathbb{R}$.

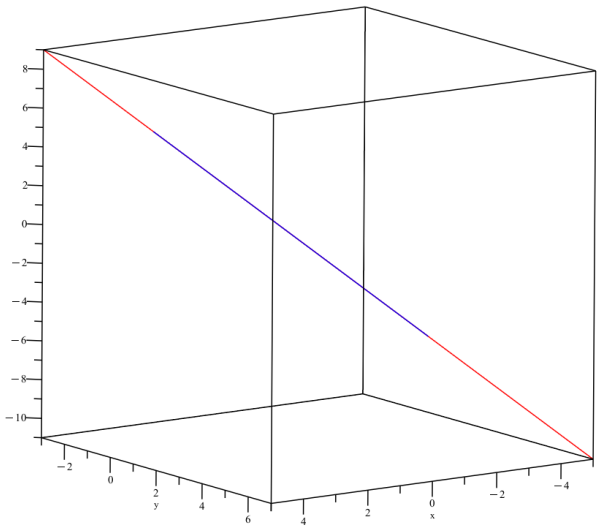
The points where the line starts also differ in the parametric form. However, once again it's because it's just an addition of some multiple of the vector. In this case, it's shifted by one vector length

$$\begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \\ 0 \end{pmatrix} - 1 \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{-1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}.$$

As demonstrated although both parametric form (c) and (d) may look different they do indeed both represent the same line. To further reinforce this fact here is the Maple graph of both parametric equations.

```
#Line·(c)
L1 := spacecurve([ −t, 2 + t, −1 − 2*t], t = −5 .. 5, color = red) :
#Line(d)
L2 := spacecurve([1/2 + (1/2) *t, 3/2 − (1/2) *t, t], t = −5 ..5, color = blue) :

display([L1, L2]);
```



f) Question (f) wants to find the vector $\vec{m} = \vec{n_1} \times \vec{n_2}$ where $\vec{n_1}$ and $\vec{n_2}$ are the normal vectors to the plane Π_1 and Π_2, respectively. Moreover, it also asks to show that the newly found

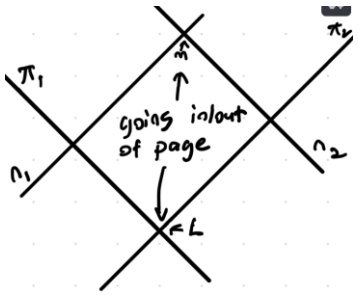
m is parallel to the line L (the intersect of the two plane). First the repeat the method in (a) for the cross product of $\overrightarrow{n_1} \times \overrightarrow{n_2}$ which is $\begin{pmatrix} -4 \\ 4 \\ -8 \end{pmatrix}$ (using Maple).

```
u := ⟨1,−5,−3⟩ :
v := ⟨−1,−3,−1⟩ :
u&xv;
```

$$\begin{bmatrix} -4 \\ 4 \\ -8 \end{bmatrix} \quad (1)$$

Then we can clearly tell that the resulting vector $\vec{m}$ is parallel to the line L. By comparing $\vec{m}$ to the direction vector from parts (c) and (d), we find it's a scalar multiple. Specifically, $\vec{m}$ is four times the direction vector from part (d), and negative eight times the direction vector from part (c).

g) Question g asks to give a geometric explanation for the result found in (f). A simple explanation is that because both line L and m is orthogonal to both normal vectors $\overrightarrow{n_1}$ and $\overrightarrow{n_2}$ they must lie in the direction common to both planes. Hence $\vec{m}$ is parallel to L.

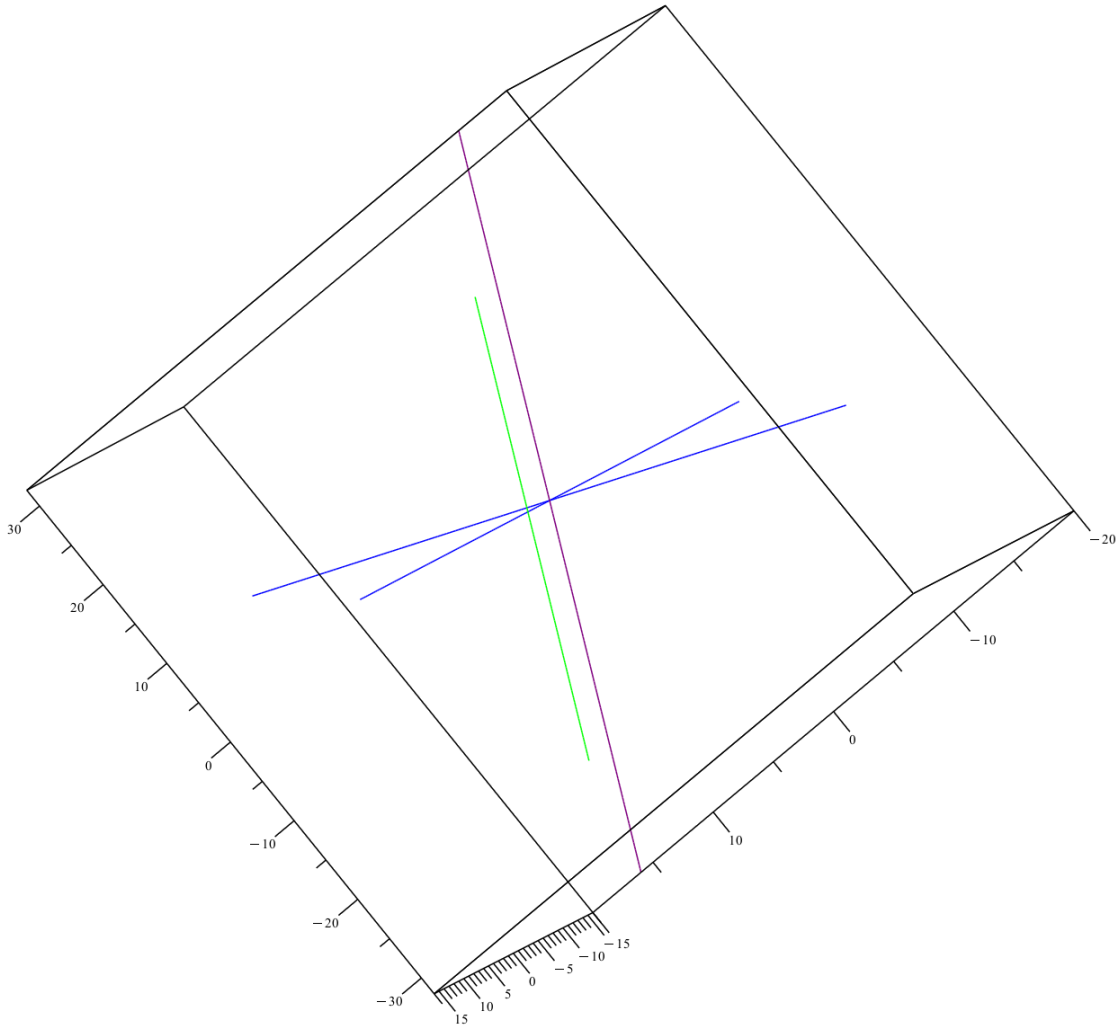


Here it is in Maple to further visualise this concept. The blue lines are $\overrightarrow{n_1}$ and $\overrightarrow{n_2}$, purple is $\vec{m}$ and green is L. The vectors are set to start from origin apart from L which is where the planes would intersect.

```
n1 := spacecurve([t, −5*t, −3*t], t=−4..4, color=blue) :
n2 := spacecurve([−1*t, −3*t, −1*t], t=−4..4, color=blue) :
m := spacecurve([−4*t, 4*t, −8*t], t=−4..4, color=purple) :

L := spacecurve([0 + (−1)*t, 2 + 1*t, −1 + (−2)*t], t=−10..10, color=green) :

display([n1, n2, m, L]);
```



Question Two

Part 1

We are also told that the instantaneous power used is:

$$P = \frac{dE}{dt}.$$

Where P is instantaneous power measured in watts, E is energy measured in joules and t is time measured in seconds.

We are also given the expression for the energy released while he watches K-drama that is

In light: $0.91(e^{-t} + t)$,

In heat: $0.55(e^{-t} + t)$,

Where t is measured in seconds.

a) This question asks us to find the combined instantaneous power emitted by the laptop, 1 hours and 39 minutes into his binge-watching session. Assuming he starts watching at $t = 0$.

First, we need to convert the time into seconds as its the units required for the formula $60 \times (39 + 60) = 5940$ seconds.

We then need to find the derivative for the energy released against time as that is equivalent to the instantaneous power released given previously

$$\begin{aligned} P &= \frac{dE}{dt} \\ &= \frac{d}{dt}\left(0.55(e^{-t} + t) + 0.91(e^{-t} + t)\right) \\ &= \frac{d}{dt}(0.55e^{-t} + 0.55t + 0.91e^{-t} + 0.91t) \\ &= \frac{d}{dt}(1.46e^{-t} + 1.46t) \\ &= -1.46e^{-t} + 1.46. \end{aligned}$$

Then subbing in $t = 5940s$ into the formula above will give the combined instantaneous power emitted by the laptop

$$P = -1.46e^{-5940} + 1.46 \approx 1.46.$$

b) This asks us to use the epsilon-M definition of limits to show that the rounded value found in (1a) is the limit of the combined instantaneous power emitted by the laptop as time goes to infinity

$$\lim_{t\to\infty} -1.46e^{-t} + 1.46 = 1.46.$$

Let's say that $P(t) = -1.46e^{-t} + 1.46$, $L = 1.46$ and $\epsilon > 0$.

$$\begin{aligned} |P(t) - L| &= |-1.46e^{-t} + 1.46 - 1.46| \\ &= 1.46e^{-t} \\ &< 1.46e^{-t}. \end{aligned}$$

In summary,

$$|P(t) - L| < 1.46e^{-t}.$$

This is our upper bound for $|P(t) - L|$.] Now

$$1.46e^{-t} < \epsilon.$$

This is equivalent to

$$t > -\ln\left(\frac{\epsilon}{1.46}\right).$$

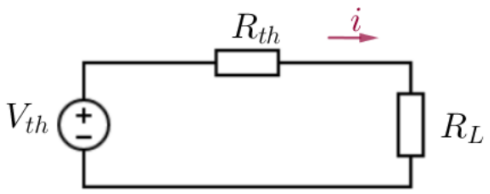
Hence whenever $t > -\ln\left(\frac{\epsilon}{1.46}\right)$ set

$$M = -\ln\left(\frac{\epsilon}{1.46}\right).$$

Then $|P(t) - L| < \epsilon$ whenever $t > M$, so $\lim_{t\to\infty} P(t) = 1.46$.

Part 2

After dinner, he needs to charge his laptop to continue watching his K-drama after dinner. His Home-made circuit is depicted below:



My home-made circuit

Where $V_{th} = 128$ volts, $R_{th} = 28\ \Omega$ and R_L is the resistance in the laptop.

The power transferred to the laptop is by the circuit is given by

$$P(R_L) = R_L\left(\frac{V_{th}}{R_{th} + R_L}\right)^2.$$

It states that the circuit is designed to delivers the maximum power to the laptop. Additionally, it states that due to technical specifications, the resistance through the laptop (R_L) cannot exceed $1\ \times 10^6\ \Omega$. We are then asked to find the resistance of the laptop.

To find the resistance of the laptop we can apply the Max-Min Theorem, since we need to find the maximum power delivered. However, before doing so we must check it the conditions are meet to apply this theorem.

First condition is that the function is on closed and bounded intervals which is obvious since you can't have negative resistance so $R_L > 0$ and it explicitly states that R_L cannot exceed $1\ \times 10^6\ \Omega$. Hence the function is bounded between $[0, 1\ \times 10^6]$.

The final condition that needs to be meet is that the functions is continuous. We can tell that it will be continuous because the denominator of rational function within the brackets will always be positive since both $R_{th} = 28\ \Omega$ is positive and $R_L > 0$. Then as the rest is elementary functions $P(R_L)$ will be continuous from $[0, 1\ \times 10^6]$.

Now we know that we can apply the max-min theorem we can find the derivative of $P(R_L)$ then set it equal to zero and see if it's a maximum

$$\begin{aligned}
\frac{dP}{dR_L} &= \frac{d}{dR_L}\left(R_L \cdot \left(\frac{V_{th}}{R_{th} + R_L}\right)^2\right) \\
&= \frac{d}{dR_L}R_L \cdot \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 + \frac{d}{dR_L}\left(\frac{V_{th}}{R_{th} + R_L}\right)^2 \cdot R_L \\
&= 1 \cdot \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 + 2 \cdot \left(\frac{V_{th}}{R_{th} + R_L}\right) \cdot \frac{-V_{th}}{(R_{th} + R_L)^2} \cdot R_L \\
&= \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 + \frac{-2{V_{th}}^2 R_L}{(R_{th} + R_L)^3} \\
&= \frac{{V_{th}}^2 \cdot (R_{th} - R_L)}{(R_{th} + R_L)^3}
\end{aligned}$$

where $R_L \in \mathbb{R}$.

Subbing in values for $V_{th} = 128$ and $R_{th} = 28\ \Omega$

$$R_L = 28\ \Omega.$$

Using maple to verify give the same answer:

$$u := x \cdot \left(\frac{128}{28 + x} \right)^2;$$

$$u := \frac{16384\,x}{(28 + x)^2} \tag{1}$$

$$v := diff(u, x);$$

$$v := \frac{16384}{(28 + x)^2} - \frac{32768\,x}{(28 + x)^3} \tag{2}$$

$$simplify(v)$$

$$\frac{458752 - 16384\,x}{(28 + x)^3} \tag{3}$$

$$solve(v = 0, x);$$

$$28 \tag{4}$$

Graphing the function $P(R_L)$ will also help visualize the answer:

$$plot(u, x = 0..100);$$

